Ioanna Vavelidou

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https://ioannavav.github.io/

EDUCATION

PhD in Electrical Engineering *Sept 2021 - Present*
Stanford University, CA
Advisor: Caroline Trippel, High Assurance Computer Architectures Lab

MS in Electrical Engineering *Sept 2022 - June 2024*
Stanford University, CA

Diploma in Electrical and Computer Engineering (Joint BSc & MEng) *Oct 2015 - July 2021*
National Technical University of Athens, Greece
Advisor: Professor Nectarios Koziris, Computing Systems Laboratory
Thesis: *Efficient Shared Resource Management in Multiprocessors using Reinforcement Learning techniques*
GPA: 9.26/10 (top 3%)
Department Entrance Rank: 1st (Highest score in the National University Admission Exams, top 0.01% nation-wide)

RESEARCH EXPERIENCE

Ph.D Research Assistant *April 2022 - Present*
Stanford University, CA
Advised by Professor Caroline Trippel
Topic: *Reliable Datacenter Infrastructure Through Defect Detection and Mitigation*

Undergraduate Researcher *Dec 2019 - July 2021*
National Technical University of Athens, Greece
Thesis: *Efficient Shared Resource Management in Multiprocessors using Reinforcement Learning techniques*

INTERNSHIPS

Student Researcher at Google Platforms *Feb 2026 - June 2026*
Developing an ML-driven probability model for predicting insidious hardware failures in Google's AI chip fleet, enabling smarter testing and performance trade-off decisions.

Student Researcher at Google DeepMind *June 2025 - Dec 2025*
Deployed and evaluated real-world ML workloads as hardware reliability tests, identifying which workloads best expose defect-induced errors in AI accelerators.

Student Researcher at Google Platforms *October 2023 - June 2024*
Designed compiler-based functional tests for Silent Data Corruption detection in CPUs.

PhD Software Engineer Intern at Meta Platforms *June 2022 - Sept 2022*
Optimized memory usage for ML workloads to improve performance and scalability

HONORS & AWARDS

Award by the President of the Republic of Greece
For achieving the highest National University Admission score in my educational district (Attica district A')
Award and Grant by the National Technical University of Athens
For achieving the highest entrance score in my department
Award and Grant by Eurobank EFG, "The Great Moment for Education"
For achieving the highest National University Admission score in my High School

PUBLICATIONS

Ioanna Vavelidou, Subho Banerjee, Eric X. Liu, Mike Fuller, Subhasish Mitra, Caroline Trippel
ITHICA: Intra-Thread Instruction Checking Approach for Defect-Induced

Data Corruptions
(in submission)

Saranyu Chattopadhyay, Keerthikumara Devarajegowda, Bihan Zhao, Florian Lonsing, Brandon Alan D'Agostino, **Ioanna Vavelidou**, Vijay Deep Bhatt, Sebastian Prebeck, Wolfgang Ecker, Caroline Trippel, Clark Barrett, Subhasish Mitra
G-QED: Generalized QED Pre-silicon Verification beyond Non-Interfering Hardware Accelerators
DAC 2023, Design Automation Conference

Hang Qiu, **Ioanna Vavelidou**, Jian Li, Evgenya Pergament, Pete Warden, Sandeep Chinchali, Zain Asgar, Sachin Katti
ML-EXRAY: Visibility into ML Deployment on the Edge
MLSys 2022, Conference on Machine Learning and Systems

MENTORSHIP

CURIS (Summer 2026): Designing and mentoring a Stanford CS undergraduate research project on using open-source LLM workloads as tests for defect-induced errors.
CURIS (Summer 2025): Designed and mentored a Stanford CS undergraduate research project on hardware fault injection in ML models.
SURA Mentorship 2026: Mentoring two early-stage undergraduate researchers through Stanford Undergraduate Research Association (SURA).
WEE Mentorship (2025-now): Mentoring an undergraduate woman in EE through Stanford's Women in Electrical Engineering mentorship program.
Stanford Science Penpals (2024-now): Exchanging monthly letters with a K-12 female student to share the experience of being a STEM scientist.

SKILLS

Languages: English (highly proficient, Michigan ECPE C2 Degree, TOEFL), French (adequate working knowledge, Delf B2 Degree), Greek (native)
Programming Languages: Python, C, C++, MATLAB, x86 Assembly, LLVM IR
Relevant Coursework: Machine Learning on Embedded Systems, Reinforcement Learning, Human-Computer Interaction Research, Cloud Computing Research, Principles of Data-Intensive Systems, Performance Engineering of Computer Systems & Networks, Computer Systems Architecture **(Stanford)**, Speech and Natural Language Processing, Pattern Recognition, Computer Vision, Advanced Robotics, Advanced Topics in Computer Architecture, Logic Design of Digital Systems, Operating Systems, Networks and Circuitry Theory, VLSI System Design **(NTUA)**

GRE SCORES

GRE General Aug. 2020: Quantitative Reasoning 169/170, Verbal Reasoning 163/170, Analytical Writing 4.5/6

PROFESSIONAL MEMBERSHIPS

- ACM Member, ACM SIGARCH
- Computing Research Association - Widening Participation (CRA-WP)
- IEEE Member